Independent Study

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This Week

- Recreating LTRAG's results and improving the pipeline
- Retesting my results and many others

Problems after last week's tests

I started growing skeptical of last week's results. Particularly because COT and standard were showing similar results. After digging a little further. I came to realize that COT's few-shot examples were not connected in the pipeline, and neither was the method producing output reasoning. Standard and COT were virtually the same. I made sure to add the few-shot examples to COT to fix this issue.

Reimplementing LTRAG

After the bug mentioned above, I grew more skeptical of the way I reimplemented LTRAG. Therefore, I decided to implement their results. I decided to choose their gpt-3.5-turbo model because it's cheap, and I can compare my results to theirs. Here is what I found compared to theirs, respectively:

Mode	Accuracy (%)	Cost ($)
Standard	54.1	0.1215
COT	62.1	0.3051
LTRAG	69.8	0.5265
Total		**0.9531**

Mine (gpt-3.5-turbo)

Model	FOLIO (Accuracy %)				
	LTRAG	Standard	CoT	LINC	Logic-LM
GPT-4o	**80.77**	73.63	78.02	72.50	78.92
DeepSeek v2.5	**78.57**	74.73	76.37	-	-
Llama3.3	**78.57**	72.53	71.43	-	-
GPT-3.5-turbo	**70.88**	56.59	59.34	62.60	61.27
Gemma2	**79.67**	59.89	62.09	-	-

LTRAG

My results and theirs follow the same pattern as theirs. I assume the differences come from random deviation. The largest deviation is COT because the difference is a bit over 2%, so I'll keep an eye out. However, I don't think it's anything to worry about. Especially because FOLIO has only 182 samples. Therefore, larger deviations can be possible.

Retesting

This bug discards my previous results. Therefore, I had to retest my experiments, and I also wanted to test more models. This mass re-experimenting prompted me to find a new way to run my program. I found multithreading to be much faster at handling 182 API calls than multiprocessing. The replacement was simple, and the only mutex lock I had to do was on the Z3 solver,, and the old process pool swapped to thread pool handles everything else well.

Results

My modified results are found on my repo, and I talk more about it in the final report.

References

AI Use:

I used Claude for the following:

1. Help understanding multithreading and how it relates to multiprocessing in Python
2. Debugging the COT pipeline

All changes were made by the author